15. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Lamp black	(A) Dull grey solid and good conductor of electricity.
(ii) Coke	(B) Light powdery substance, having a velvet touch.
(iii) Gas carbon	(C) Hardest naturally occurring substance.
(iv) Graphite	(D) Greyish black porous solid.
(v) Diamond	(E) Metallic lustre and good conductor of electricity.

(a) (i) → (D); (ii) → (B); (iii) → (A); (iv) → (E); (v) → (A)
(b) (i) → (B); (ii) → (D); (iii) → (A); (iv) → (E); (v) → (C)
(c) (i) → (B); (ii) → (A); (iii) → (D); (iv) → (E); (v) → (C)
(d) (i) → (B); (ii) → (D); (iii) → (A); (iv) → (C); (v) → (E)

Space for rough work

Answer Keys

Assessment Test I

1. (b) **2.** (a) **3.** (c) **4.** (b) **5.** (c) **6.** (a) **7.** (b) **8.** (d) **9.** (a) **10.** (a)
11. (c) **12.** (c) **13.** (d) **14.** (a) **15.** (c)

Assessment Test II

1. (a) **2.** (b) **3.** (d) **4.** (a) **5.** (b) **6.** (c) **7.** (d) **8.** (b) **9.** (b) **10.** (b)
11. (d) **12.** (a) **13.** (c) **14.** (d) **15.** (a)

Assessment Test III

1. (a) **2.** (d) **3.** (c) **4.** (a) **5.** (d) **6.** (a) **7.** (a) **8.** (d) **9.** (d) **10.** (b)
11. (c) **12.** (b) **13.** (a) **14.** (b) **15.** (a)

Assessment Test IV

1. (d) **2.** (b) **3.** (d) **4.** (d) **5.** (d) **6.** (b) **7.** (c) **8.** (b) **9.** (c) **10.** (a)
11. (a) **12.** (d) **13.** (a) **14.** (c) **15.** (b)

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Some Important Elements and their Compounds 7

Reference: Coursebook - IIT Foundation Chemistry Class 8; Chapter - Some Important Elements and their Compounds; Page number - 7.1–7.36

Assessment Test I

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Arrange the following in the increasing order of temperature range in which they are obtained during heating of sulphur.
   (A) Less viscous mobile liquid　(B) Monoclinic sulphur
   (C) Sulphur vapour　(D) Viscous liquid
   (E) Pale yellow mobile liquid
   (a) BEADC　(b) BECDA
   (c) BEDAC　(d) BECAD

2. Arrange the following reactions in the order of formation of NO_2, NO, N_2O, and H_2.
   (A) Zn + HNO_3 (dil.)　(B) Cu + HNO_3 (dil.)
   (C) Cu + HNO_3 (conc.)　(D) Mg + HNO_3 (dil.)
   (a) CBAD　(b) CBDA　(c) BCAD　(d) BCDA

3. **Assertion (A):** Bleaching action of SO_2 is temporary.
   Reason (R): In bleaching action, SO_2 acts as a reducing agent.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

4. **Assertion (A):** Nitric acid is used in the purification of gold.
   Reason (R): HNO_3 is strong reducing agent.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

Space for rough work

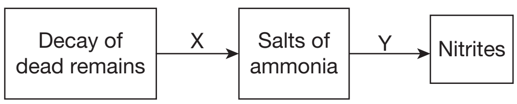
5. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Nitrogen	(A) Gunpowder
(ii) Phosphorus	(B) Storage batteries
(iii) Sulphur	(C) Rat poison
(iv) Hydrogen sulphide	(D) Refrigerant
(v) Sulphuric acid	(E) Analytical reagent

(a) (i) → (D); (ii) → (C); (iii) → (A); (iv) → (E); (v) → (B)
(b) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (B)
(c) (i) → (D); (ii) → (E); (iii) → (A); (iv) → (B); (v) → (C)
(d) (i) → (E); (ii) → (C); (iii) → (D); (iv) → (B); (v) → (A)

6. Aqueous solution of 'X' on treatment with copper sulphate solution gives a pale blue precipitate of 'Y'. What could be X and Y?
(a) X is NH_3; Y → $Cu(OH)_2$ (b) X is NH_3; Y → $CuOH$
(c) X is H_2S; Y → CuS (d) X is H_2S; Y → Cu_2S

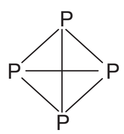
7. Identify 'X' and 'Y' in the following flowchart.



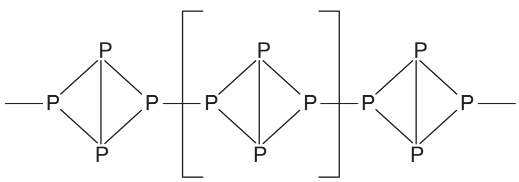
(a) X is ammonifying bacteria. Y is symbiotic bacteria.
(b) X is denitrifying bacteria. Y is *Nitrobacter* bacteria.
(c) X is ammonifying bacteria. Y is *Nitrosomonas* bacteria.
(d) X is *Nitrosomonas* bacteria. Y is *Nitrobacter* bacteria.

8. Study the diagrams given below and answer the questions.

Structure of 'X'.



Structure of 'Y'.



(A) X is red phosphorous and Y is white phosphorous.
(B) Y is used in match industry in preference to X.

(C) Both X and Y are allotropes and possess same chemical properties and different physical properties.

(D) Both X and Y are allotropes and possess same physical and chemical properties.

Identify true statements.

(a) (A) and (B) (b) (B) and (C)

(c) (A) and (C) (d) (B) and (D)

9. Identify true statements regarding rhombic sulphur and monoclinic sulphur.

(a) Both are crystalline and insoluble in CS_2.

(b) Both are amorphous and soluble in CS_2.

(c) Both are crystalline and soluble in CS_2.

(d) Both are amorphous and insoluble in CS_2.

10. Sugar, on treating with sulphuric acid, turns to a black coloured mass. The property of sulphuric acid exploited in this process is __________.

(a) Acid nature (b) Oxidizing property

(c) Dehydrating property (d) Bleaching property

11. Aqueous solutions of two metal sulphates on treatment with hydrogen sulphide separately give flesh coloured and white precipitates. What are the two metals?

(a) Mn, Zn (b) Pb, Cu

(c) Pb, Mn (d) Zn, Cu

12. Identify 'X' in the following reactions.

$Cl_2 + NaOH$ (Cold and dilute) $\rightarrow X + Y + Z$

$Cl_2 + NaOH$ (Hot and concentrated) $\rightarrow X + A + Z$

(a) NaCl (b) $NaClO_4$

(c) NaOCl (d) $NaClO_3$

13. Identify 'A', 'Y' and 'Z' in the following reaction.

$B + A \xrightarrow[\text{(Sunlight)}]{} Y : Y + H_2O \rightarrow Z; Z + Zn \rightarrow ZnCl_2 + B$

(a) $A \rightarrow Cl_2$, $Y \rightarrow HCl$ $Z \rightarrow$ Hydrochloric acid

(b) $A \rightarrow Cl_2$, $Y \rightarrow HCl$ $Z \rightarrow$ Hypochlorous acid

(c) $A \rightarrow H_2$, $Y \rightarrow HCl$ $Z \rightarrow$ Hydrochloric acid

(d) $A \rightarrow Cl_2$, $Y \rightarrow HCl$ $Z \rightarrow$ Chloric acid

14. **Assertion (A):** Thermosetting plastics are used for making handles of kitchenware.

Reason (R): Thermosetting plastics are made up of cross linked polymeric chains.

Space for rough work

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

15. In olden days, white phosphorus was used in safety matches. Now, it has been replaced by red phosphorus. Identify the property due to which the replacement has been made.

(a) Red phosphorus being red in colour evolves more light and heat energy.
(b) White phosphorus has a very low ignition temperature than red phosphorus.
(c) Red phosphorus has isolated P_4 tetrahedral units under normal conditions and hence is not poisonous in nature.
(d) White phosphorus sublimes under normal conditions and hence, the material is wasted.

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Action of heat on sulphur is described below.
   (i) Lambda sulphur $\rightarrow S_{\mu}$ - dark.
   (ii) Rhombic sulphur $\rightarrow$ Monoclinic sulphur.
   (iii) Long chain of sulphur atoms $\rightarrow S_2$ and S_4 units.
   (iv) Monoclinic sulphur $\rightarrow$ Lambda sulphur.
   (v) S_2 and S_4 units $\rightarrow$ Atoms of sulphur.
   Arrange the following temperatures in the order of statements given below.
   (A) 114°C (B) 95.6°C
   (C) 445°C (D) 160°C
   (E) 200°C
   (a) DBAEC (b) DBEAC
   (c) DEBAC (d) DEABC

2. Arrange the following reactions involved in Brown ring test for metal nitrate in the correct order.
   (A) $2HNO_3 \rightarrow H_2O + 2NO + 3[O]$
   (B) $NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$
   (C) $FeSO_4 + NO \rightarrow FeSO_4.NO$
   (D) $NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$
   (a) BDAC (b) BDCA
   (c) DBAC (d) DBCA

3. **Assertion (A):** SO_2 is used in the sugar industry.
   Reason (R): SO_2 decolourizes sugarcane juice due to its bleaching action.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

4. **Assertion (A):** Metals such as iron, cobalt, etc., do not react with HNO_3.
   Reason (R): Fe and Co become passive on treatment with HNO_3.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

Space for rough work

5. Match the statements of Column A with those of Column B.

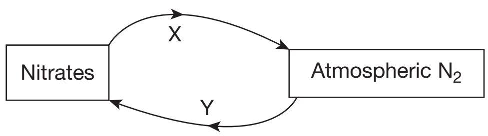
Column A	Column B
(i) Nitrogen	(A) Yellow colour substance
(ii) White Phosphorus	(B) Rotten egg smell
(iii) Sulphur	(C) Corrosive in nature
(iv) Hydrogen sulphide	(D) Unreactive substance
(v) Sulphuric acid	(E) Garlic smell

(a) (i) → (D); (ii) → (C); (iii) → (A); (iv) → (E); (v) → (B)
(b) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (B)
(c) (i) → (D); (ii) → (E); (iii) → (A); (iv) → (B); (v) → (C)
(d) (i) → (E); (ii) → (C); (iii) → (D); (iv) → (B); (v) → (A)

6. Copper sulphate on treatment with excess aqueous solution of 'X' gives a dark blue solution of 'Y'. Identify X and Y.

(a) $X \rightarrow NH_3$; $Y \rightarrow [Cu(NH_3)_2]SO_3$
(b) $X \rightarrow NH_3$; $Y \rightarrow [Cu(NH_3)_2]SO_4$
(c) $X \rightarrow (NH_4)_2SO_4$; $Y \rightarrow [Cu(NH_3)_2]SO_3$
(d) $X \rightarrow (NH_4)_2SO_4$; $Y \rightarrow [Cu(NH_3)_2]SO_4$

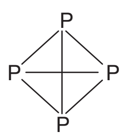
7. Identify 'X' and 'Y' in the following figure.



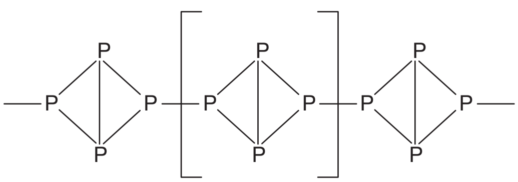
(a) X is ammonifying bacteria. Y is *Nitrobacter* bacteria.
(b) X is denitrifying bacteria. Y is symbiotic bacteria.
(c) X is ammonifying bacteria. Y is *Nitrosomonas* bacteria.
(d) X is *Nitrosomonas* bacteria. Y is *Nitrobacter* bacteria.

8. Study the diagram given below and identify the wrong statements.

Structure of 'X'.



Structure of 'Y'.



Space for rough work

(A) X is poisonous and Y sublimes at 565 K.

(B) X is a brittle powder and Y possesses the smell.

(C) X gradually changes its colour. Y exists in dark of garlic colour.

(D) X is amorphous and Y is crystalline.

(a) (A) and (B) (b) (B) and (C)

(c) (B) and (D) (d) (A), (C), and (D)

9. Identify the true statements regarding alpha sulphur (X), beta sulphur (Y), and gamma sulphur (Z).

(A) X $\rightarrow$ Rhombic sulphur; Y $\rightarrow$ Prismatic sulphur; Z $\rightarrow$ Plastic sulphur

(B) Both X and Y are more stable at normal temperature.

(C) X possesses octahedral shape and Y has a needle shape.

(D) Z does not have a sharp melting point.

(a) ABC (b) ACD (c) BCD (d) ABD

10. In which among the following reactions does H_2SO_4 show oxidizing property?

(a) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

(b) $C + 2H_2SO_4 \rightarrow 2H_2O + 2SO_2 + CO_2$

(c) $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$

(d) $CuSO_4.5H_2O \rightarrow CuSO_4 + 5H_2O$

11. Hydrogen sulphide, on combustion, gives sulphur and water. This reaction indicates the ______ of hydrogen sulphide.

(a) acidic nature (b) basic nature

(c) reducing property (d) oxidizing property

12. Manganese dioxide, on treatment with conc. HCl, gives gas ‘X’. Identify the false statement regarding X.

(a) It is a good bleaching agent and an oxidizing agent.

(b) It is collected by the downward displacement of water.

(c) It is used in the manufacture of bleaching powder.

(d) It is greenish yellow in colour.

13. Identify the reaction which correctly represents the laboratory method of preparation of hydrogen chloride gas.

(a) $H_2 + Cl_2 \xrightarrow{\text{(Sunlight)}} 2HCl$

(b) $2HI + HgCl_2 \longrightarrow 2HCl + HgI_2$

(c) $NaHSO_4 + NaCl \xrightarrow{> 200^\circ C} (> 200^\circ C)\ Na_2SO_4 + HCl$

(d) $NaCl + H_2SO_4 \xrightarrow{200^\circ C} NaHSO_4 + HCl$

14. **Assertion (A):** Bakelite is a plastic which cannot be remoulded.

Reason (R): Bakelite is a thermoplastic polymer.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

15. 'X' and 'Y' are two allotropic forms of phosphorus and Y is used preferably in match industry. Which of the following statements is true regarding the two samples X and Y?

(a) 'X' has greater density than 'Y'.
(b) In X, P_4 units are linked by P-P bonds.
(c) In Y, P_4 units are linked by P-P bonds.
(d) Both of them undergo sublimation.

Space for rough work